Name & Surname **Polimi ID**

Exercise 1

A. Please describe the parallel gather pattern, including assumptions, complexity, and examples. (1)

B. Please describe how the Split parallel pattern works. How is it implemented (pseudo code required)? (2)

C. How and under what conditions can the following code be parallelized? Is there a way to write the code for better parallelization? (1)

```
for (j=10; j<100; j++)
        for (i=1; i<100; i++)
            a[0][j] += f(a[i][j-10]);
```

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Exercise 2

A. Is DRAM burst affecting the corner-turning optimization? (1) **This question can be skipped if you got 1 or 3 points in the second challenge.**

B. Please briefly describe what DRAM Banking is meant for. How can CUDA take advantage of DRAM banking? (2) **This question can be skipped if you got 2 or 3 points in the second challenge.**

C. Please describe the sequential consistency memory model. (1)

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Exercise 3

A. Consider the following CUDA code fragment:

```
unsigned int t = threadIdx.x;
unsigned int start = 2*blockIdx.x*blockDim.x;
partialSum[t] = input[start + t];
partialSum[blockDim.x+t] = input[start+ blockDim.x+t];

for (unsigned int stride = 1; stride <= blockDim.x; stride *= 2)
{
    __syncthreads();
    if (t % stride == 0) {partialSum[2*t]+= partialSum[2*t+stride];}
}
```

If the block size is 1024 and the warp size is 32, how many warps in a block will have divergence during the iteration where the stride is equal to 64? (1.5)

B. Assume that each CUDA atomic operation takes 4ns to complete in L2 cache and 100ns to complete in DRAM. If 90% of atomic operations in a program hit in L2 cache, what is the approximate throughput for atomic operations on the same global memory variable? (1.5)

C. Is it possible to launch multiple different CUDA kernels in parallel? How? (1)

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Exercise 4

A. Describe the effect of the following OpenMP pragma: (2)

```
#pragma omp target map(a,b)
```

B. A Domain-Specific Language provides productivity and performance at the expense of generality. Is this statement true or false? Explain. (1)

C. Describe the ASAP scheduling algorithm. (1)